AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 5 · STUDENT LESSON

Pictures Become Numbers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 2: stored input and its interpretation are different stages.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Encode and reconstruct a small image using an explicit grid convention. Try the recall before opening notes.

Your learning target

I can encode and reconstruct a small image using an explicit **grid** convention.

I can explain why **pixel** counts alone cannot recover an arrangement.

I can distinguish numerical image data from a model's **interpretation** of that data.

Remember before reading

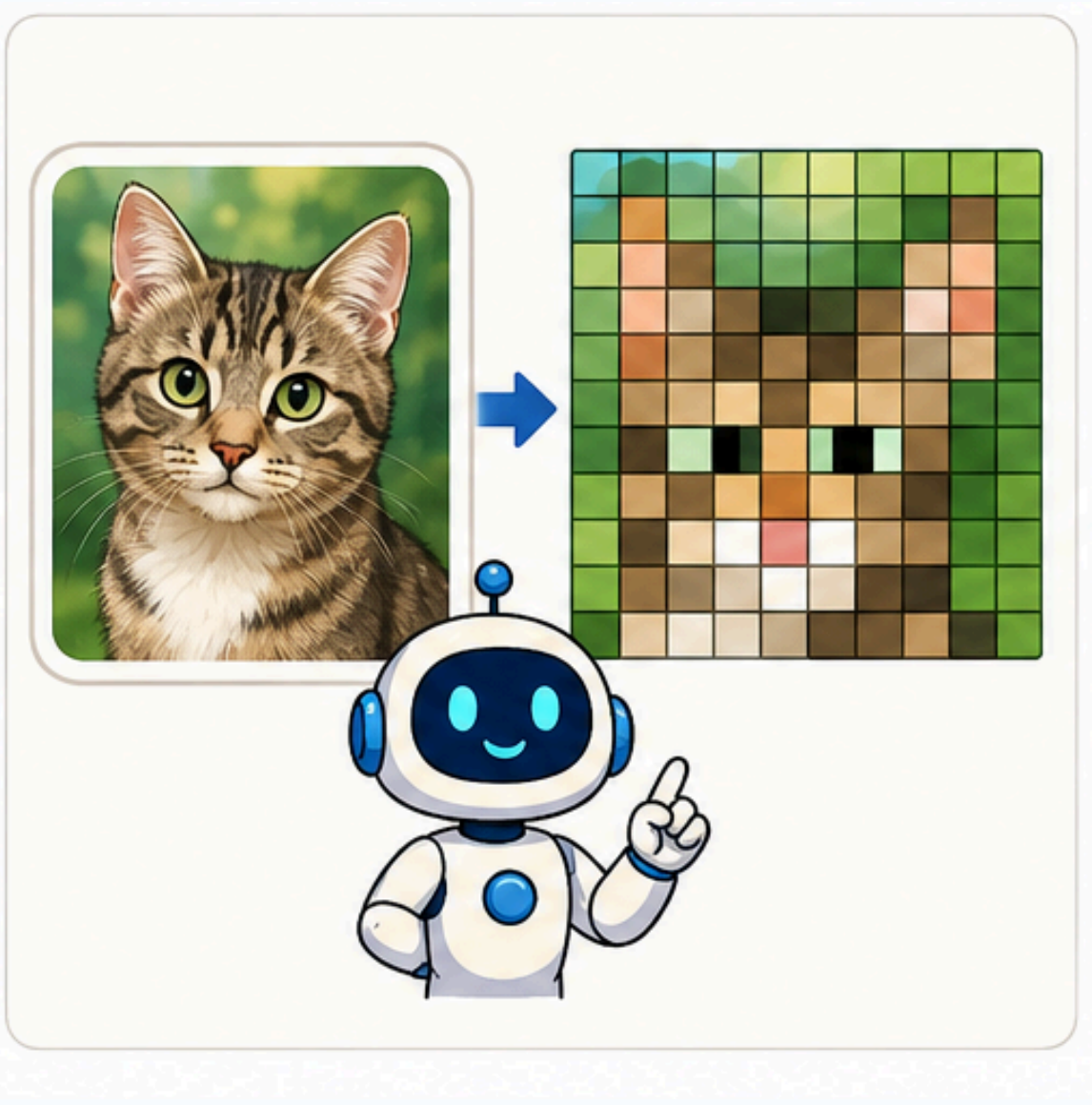
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A digital image is a **grid** of measured color values.



Your target

LEARNING TARGET Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first.

> Explain why the image changes although the six values are identical.

Learner name:

Readiness for the connected explanation

> Two thresholds can agree on all **fitting** examples but disagree on a new **input**. Does zero **fitting** error identify one uniquely correct rule?

This week's end goal

By the end, I can encode **pixel** positions and show why a count cannot reconstruct a picture.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Pictures Become Numbers case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A 5 by 5 black-and-white picture can be stored as 25 numbers, one value for each **pixel** position.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Imagine sending a small black-and-white sign using only numbers. We can divide the sign into a **grid**, record 1 for a black square and 0 for a white square, and read each row from left to right, starting at the top.

The receiver needs the width, height and these conventions to reconstruct the same arrangement.

A **pixel** is a position with a recorded value. In our simplified binary image, one value selects black or white. Other image formats can record brightness levels or multiple colour **channel**s.

The convention 1 means black belongs to this exercise; it is not a universal rule for every digital image.

Encoding a picture does not recognise what it depicts. A program can store or display the numbers without calling the image a letter, an arrow or a person.

Interpretation is another operation, just as displaying a book title was different from reading the entire book in Week 2.

STUDENT LESSON · 3 OF 18

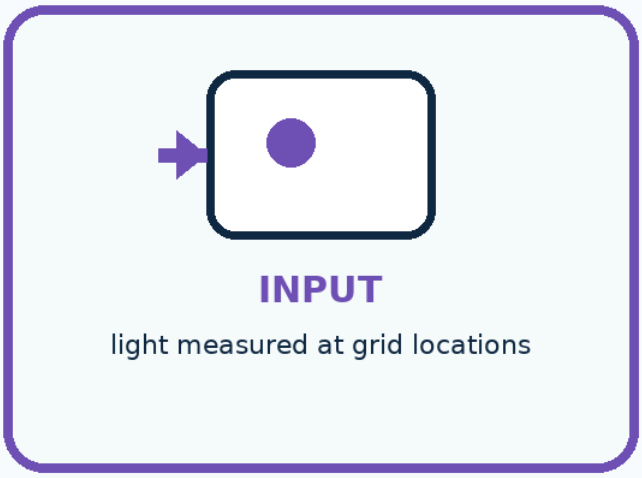
See how it works

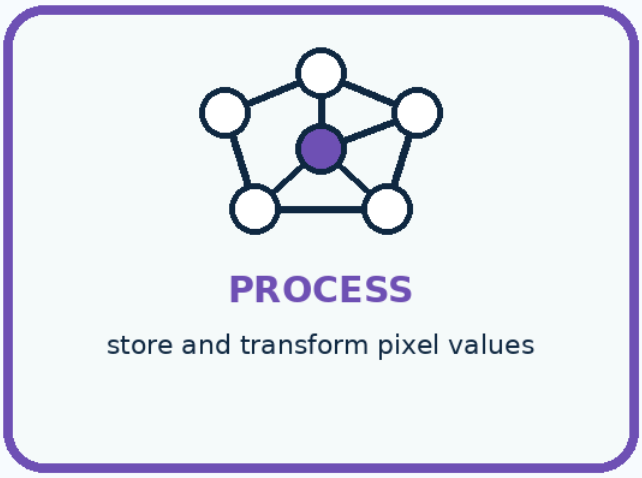
> Follow the mechanism for Pictures Become Numbers. Connect each labelled input, deciding step and result.

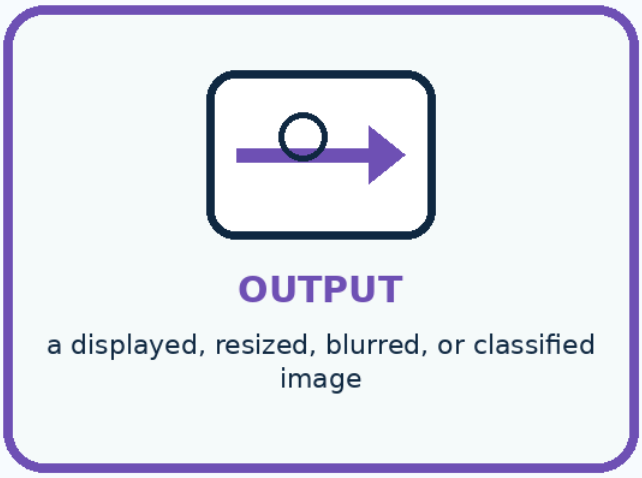
VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT light measured at **grid** locations

PROCESS store and transform **pixel** values

OUTPUT a displayed, resized, blurred, or classified image

Key idea

CORE IDEA A digital image is a **grid** of measured color values.

How the mechanism works

UNDER THE HOOD Digital images sample a scene across a spatial **grid**. Each **pixel** stores **channel** values, commonly red, green, and blue.

Resolution controls sampling density; transformations operate on numbers, not on an intact object inside the computer.

Draw a fresh example with all three stages:

A closer explanation

A three by three binary image

Read rows top to bottom. Within each row read left to right. 1 is black and 0 is white.

0	1	0
1	1	1
0	1	0

The image has width 3 and height 3, so it contains 9 pixels. Five are black. The row records preserve where those five black pixels occur.

The first row is 0,1,0, so only its middle **pixel** is black. The second row is 1,1,1, so all three are black. The third row repeats the first. This reconstructs the displayed cross-like pattern under our convention.

A flattened record would be 0,1,0,1,1,1,0,1,0. Without the **grid** dimensions, the same nine values could also be drawn in one row of nine. Keep the shape information with the values.

Width times height counts **pixel**s; it does not tell us their contents.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

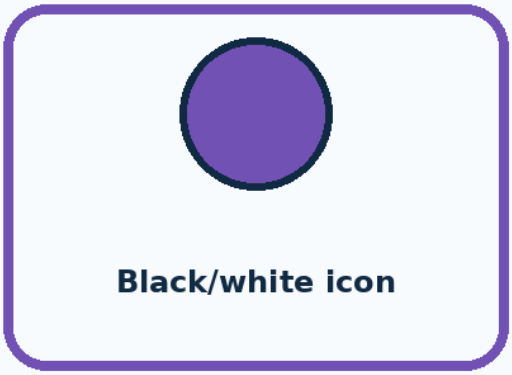
Compare the cases

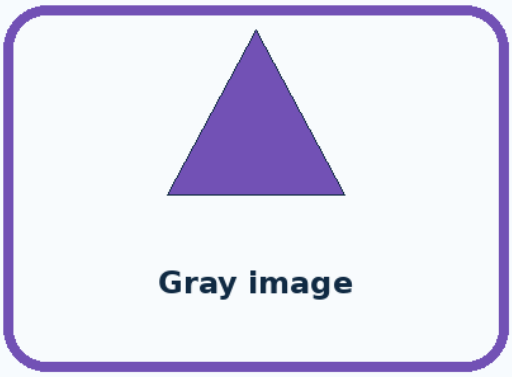
> Compare the supplied cases for Pictures Become Numbers. Identify the detail that changes the conclusion.

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Black/white icon Clue: 0 and 1 values. Meaning: two possible brightness levels.

The case to investigate

CASE 2 — Gray image Clue: one brightness number per **pixel**. Meaning: many light levels.

The case to investigate

CASE 3 — Color image Clue: red, green, blue values. Meaning: three **channel**s.

The case to investigate

CASE 4 — Blur Clue: nearby values are mixed. Meaning: detail is reduced.

Keep this boundary clear

BOUNDARY A **pixel** is a measurement at a location, not a tiny copy of the object.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Change A's bottom-left **pixel** from 0 to 1.

Predicted result

Prediction: A now has four dark **pixel**s; its row **encoding** is 101 / 110.

Why the result follows

Reason: Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

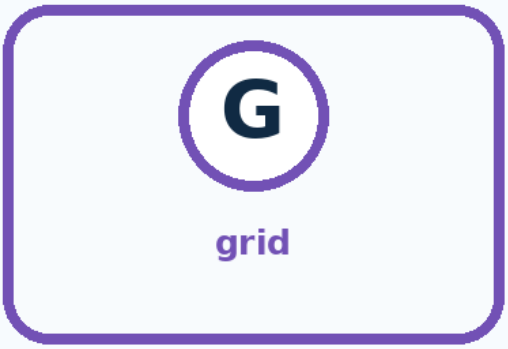
> Use the vocabulary of Pictures Become Numbers to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
pixel	one picture element in a grid	
grid	rows and columns that locate values	

Word	Meaning	My example
channel	one measured color component	
resolution	number of pixels used to represent an image	

Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR The computer stores a real cat inside the image file.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Encoding	A convention for representing information with symbols or numbers
Spatial arrangement	Where elements are located relative to one another
Interpretation	A meaning or category assigned to a representation using a rule, model or human judgment

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STUDENT LESSON · 6 OF 18

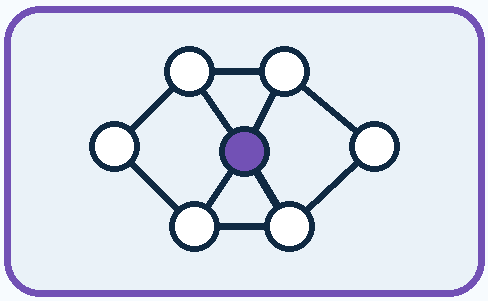
Follow a worked example

> Follow the worked trace for Pictures Become Numbers, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A 5 by 5 black-and-white picture can be stored as 25 numbers, one value for each **pixel** position.

Step 1

STEP 1 — OBSERVE light measured at **grid** locations

Step 2

STEP 2 — TRACE store and transform **pixel** values

Step 3

STEP 3 — CONCLUDE a displayed, resized, blurred, or classified image

Model answer

MODEL REASONING Each **grid** position becomes one displayed square. A blur combines nearby values, so isolated corners and thin lines disappear before large regions.

The computer transforms numbers; the human viewer recognizes the shape.

A claim to avoid

DO NOT OVERCLAIM A **pixel** is a measurement at a location, not a tiny copy of the object.

What new **evidence** could change this answer?

A closer explanation

Worked explanation · Pictures Become Numbers

Suppose we discard the row records and retain only black **pixel** count=5. That count matches the cross, but it also matches rows 1,1,1 then 1,1,0 then 0,0,0. The second arrangement fills the top row and two positions below it.

Both have five black **pixel**s and nine **pixel**s in total, yet they are different images.

Stored information	What it establishes	What remains unknown
Width 3 and height 3	There are 9 positions	The value at each position
Black pixel count 5	Five positions are black under this code	Which five positions
Dimensions and all three ordered rows	The full binary grid under this convention	What a viewer or model will call the picture

A **pixel** count can still be a useful feature for some tasks. For example, it can separate empty **grid**s from **grid**s with at least one mark. But it cannot separate every shape.

To judge a feature, ask whether it retains the distinctions required by the task.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Pictures Become Numbers

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.

The facts we will use

> **Supplied facts**
>
> Dark=1, light=0; rows are read top to bottom, left to right.

> **Supplied facts**
>
> Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

AI ACADEMY / AI-1/5

Pixel positions: two badges, same dark count

	Left	Middle	Right
A top	1	0	1
A bottom	0	1	0
B top	1	1	1
B bottom	0	0	0

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/5

Pictures Become Numbers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Pixel convention

Each image has two rows and three columns; 1 means dark.

2. Badge A

Top 101, bottom 010: three dark pixels.

3. Badge B

Top 111, bottom 000: also three dark pixels.

4. Compare arrangements

The totals agree, but middle-top and middle-bottom positions differ.

CHANGE ONE THING

Change A's bottom-left pixel from 0 to 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Pixel convention	Each image has two rows and three columns; 1 means dark.
2. Badge A	Top 101, bottom 010: three dark pixels.
3. Badge B	Top 111, bottom 000: also three dark pixels.
4. Compare arrangements	The totals agree, but middle-top and middle-bottom positions differ.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **grid** preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges.

Dimensions and reading order belong to the representation, not to an optional caption.

What this example does not establish

Keep the stated limit

This toy binary **encoding** does not include real photographs' colour **channel**s, compression or illumination effects.

Keep the stated limit

Read the labelled picture: Find a position where the original badges differ, then read the changed A back as a picture.

Keep the stated limit

Change one thing: Change A's bottom-left **pixel** from 0 to 1. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Pictures Become Numbers claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Black/white icon	0 and 1 values	two possible brightness levels
Gray image	one brightness number per pixel	many light levels
Color image	red, green, blue values	three channels
Blur	nearby values are mixed	detail is reduced

Repair two tempting claims

Challenge claim A

CLAIM A The computer stores a real cat inside the image file.

Repair:

Challenge claim B

CLAIM B More **pixel**s always guarantee a correct classification.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

Higher **resolution** offers more positions for recording detail, but it does not guarantee a useful image. A larger blurry image can still hide a feature, and a cropped image can omit it entirely.

A model's label also depends on what it was fitted to distinguish and the conditions represented in its examples.

Keep claims at the right stage. These nine values encode this **grid** is a reconstruction claim. This **grid** depicts a plus sign is an **interpretation**. This sign tells a person what to do adds context and authority.

One does not automatically establish the others.

Can two different three by three images have the same black-pixel count? Explain with a specific arrangement change

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STUDENT LESSON · 8 OF 18

Practise together

Practise Pictures Become Numbers with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A digital image is a **grid** of measured color values.

Use a fresh example to explain **pixel**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: More **pixel**s always guarantee a correct classification.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Use width 3, height 2 and rows 1,0,1 then 0,1,0. Draw two rows of three squares and fill exactly the positions marked 1. Check by reading your drawing back into numbers.

A round trip from code to drawing to code can reveal a swapped row or a missed **pixel**.

> **Draw the two by three image, write its black-pixel count and explain why saying only three black pixels would be insufficient**

If an image classifier mislabels a picture, first inspect the **input** representation. Was the width correct? Were the rows reversed? Did the program use the intended colour convention?

A later classifier cannot reliably repair every mistake in how its **input** was prepared.

Practise with less help: Preserve the missing field

Use workbook W2 for the connected example “Images as numerical **grid**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Images as numerical **grid**s. Proposed explanation: “The number one always means black”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Find a position where the original badges differ, then read the changed A back as a picture.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

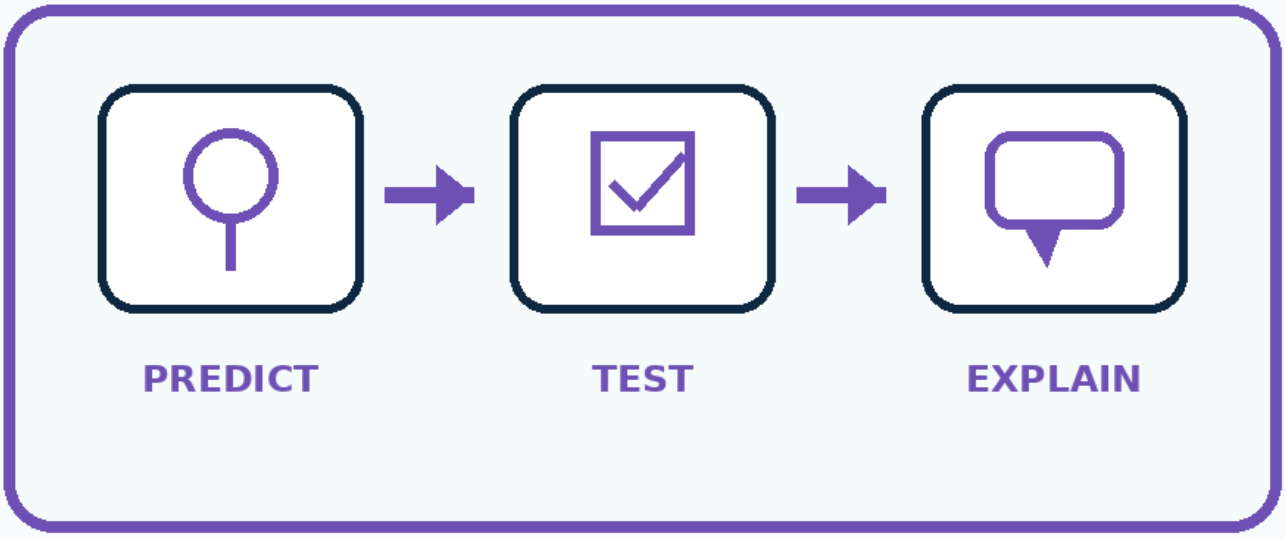
Run the investigation

> Investigate Pictures Become Numbers. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Pixel Picture Decoder

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Decode a tiny **pixel** picture and predict a blurred version.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

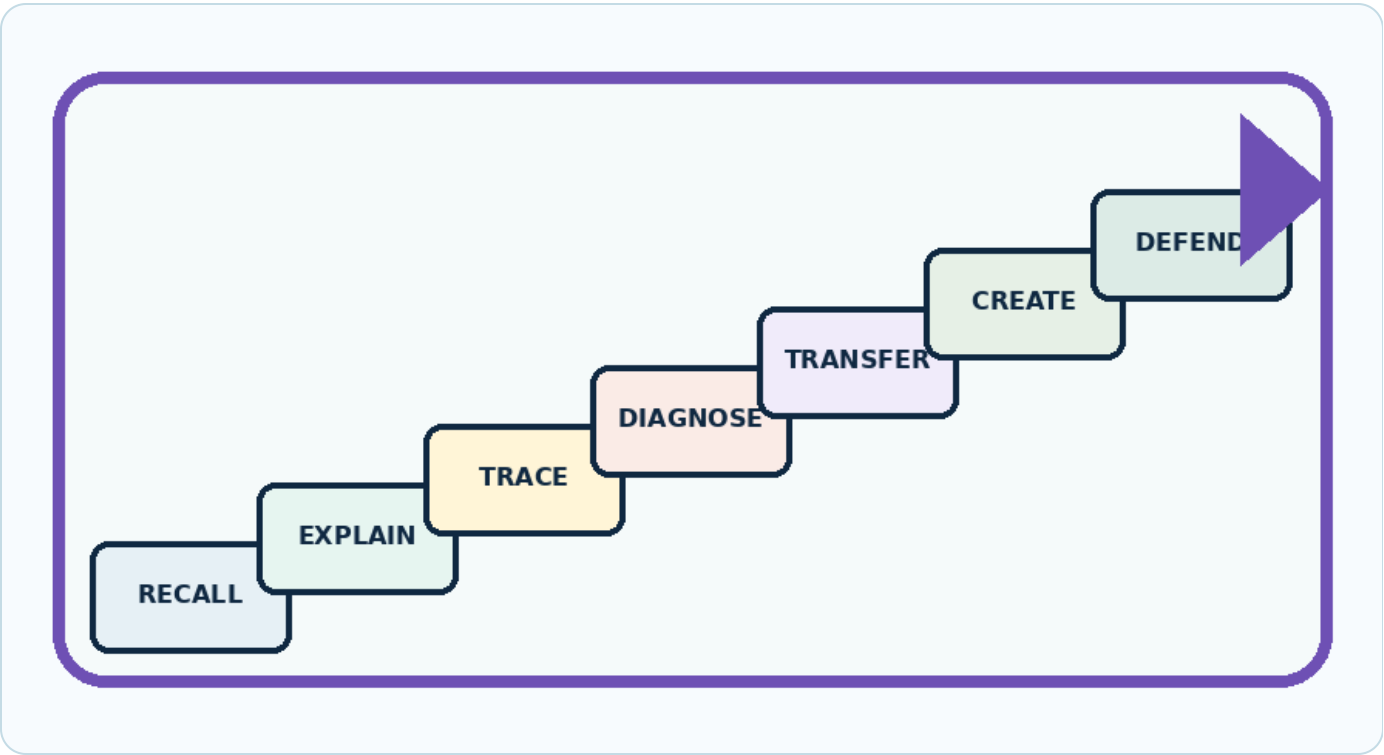
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Pictures Become Numbers. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **pixel**.

EXPLAIN A digital image is a **grid** of measured color values.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: The computer stores a real cat inside the image file.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

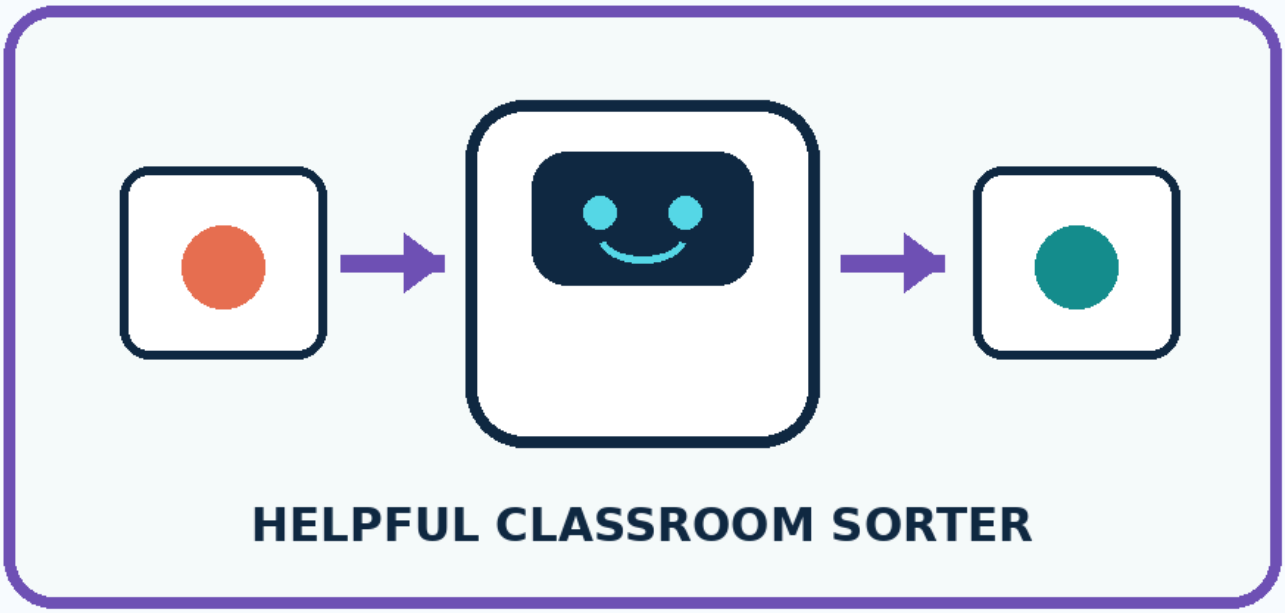
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



> THIS WEEK'S CHECKPOINT Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

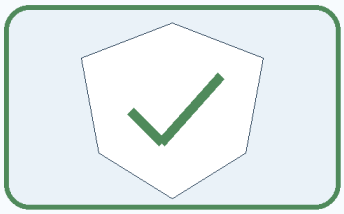
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN A **pixel** is a measurement at a location, not a tiny copy of the object.

My project decision:

Evidence for your project

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape.

STUDENT LESSON · 12 OF 18

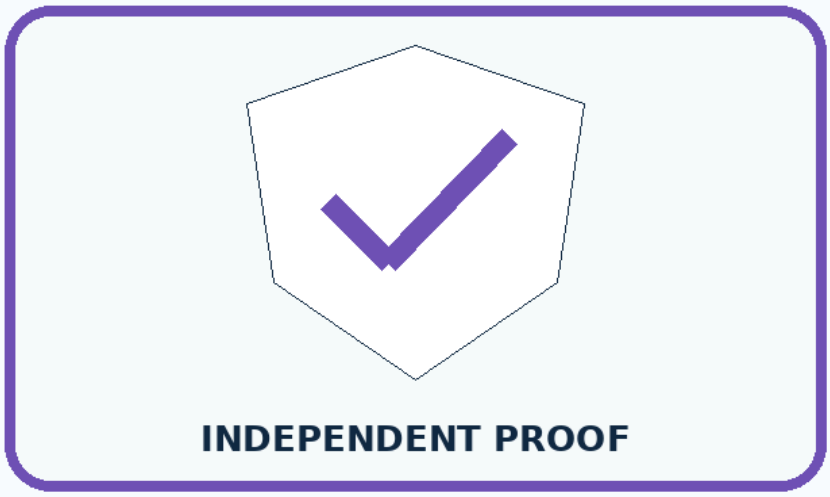
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Decode a 4 by 4 number **grid**, identify the pictured shape, then predict

which details disappear when neighboring cells are averaged.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Each **grid** position becomes one displayed square. A blur combines nearby values, so isolated corners and thin lines disappear before large regions.

The computer transforms numbers; the human viewer recognizes the shape.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Sound Becomes a Signal.

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Pictures Become Numbers case for supported practice. Repeated examples do not become fresh assessments.

Week 5 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

The question

Draw it, then change the top-left value to 1.

Worked explanation

Initially top-right and bottom-left are black. The change produces three black squares. **Pixel** values encode an image; object recognition is a separate **interpretation**.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

A picture represented by four numbers

Evidence or stage	Value or result
Top left	0
Top right	1

Evidence or stage	Value or result
Bottom left	1
Bottom right	0

A 2 by 2 image has rows [0,1] and [1,0]. Zero means white and one means black.

Draw it, then change the top-left value to 1.

Initially top-right and bottom-left are black. The change produces three black squares. **Pixel** values encode an image; object recognition is a separate **interpretation**.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Design a new counterexample

Create two different images of width 4 and height 2, each with exactly three black **pixel**s. Write both complete row **encoding**s. Explain which fact their shared count retains and which difference it discards.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 5 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Use 0 for dark and 1 for light. Draw a two-by-two image with rows [1,0] and [0,1]. Then change only the bottom-left **pixel** to light. Which number changes?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Choose which visible item features the Classroom Sorter may represent as image measurements and name one feature that lighting could distort.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Encode a 2×3 monochrome image with top row black–white–black and bottom row white–black–white, using 1 for black. A partner reads columns first. Explain why the image changes although the six values are identical.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Pictures Become Numbers, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.

By the end, I can	How I will show it
Encode and reconstruct a small image using an explicit grid convention	Explain the deciding step, show a trace or test, and state one limit.
Explain why pixel counts alone cannot recover an arrangement	Explain the deciding step, show a trace or test, and state one limit.
Distinguish numerical image data from a model's interpretation of that data	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Pictures Become Numbers without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Encode and reconstruct a small image using an explicit **grid** convention

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The totals agree, but middle-top and middle-bottom positions differ.

1. Compare arrangements

2. Badge B

3. **Pixel** convention

4. Badge A

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare arrangements The totals agree, but middle-top and middle-bottom positions differ. A **grid** preserves where each brightness belongs.

Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges. Dimensions and reading order belong to the representation, not to an optional caption.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change A's bottom-left **pixel** from 0 to 1. Which conclusion is supported by the supplied facts?

1. A now has four dark **pixel**s; its row **encoding** is 101 / 110.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A now has four dark **pixel**s; its row **encoding** is 101 / 110. Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

Repair a missing building block

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used labels and which stage used only the new measurement..

Return to this week and explain what you changed.

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

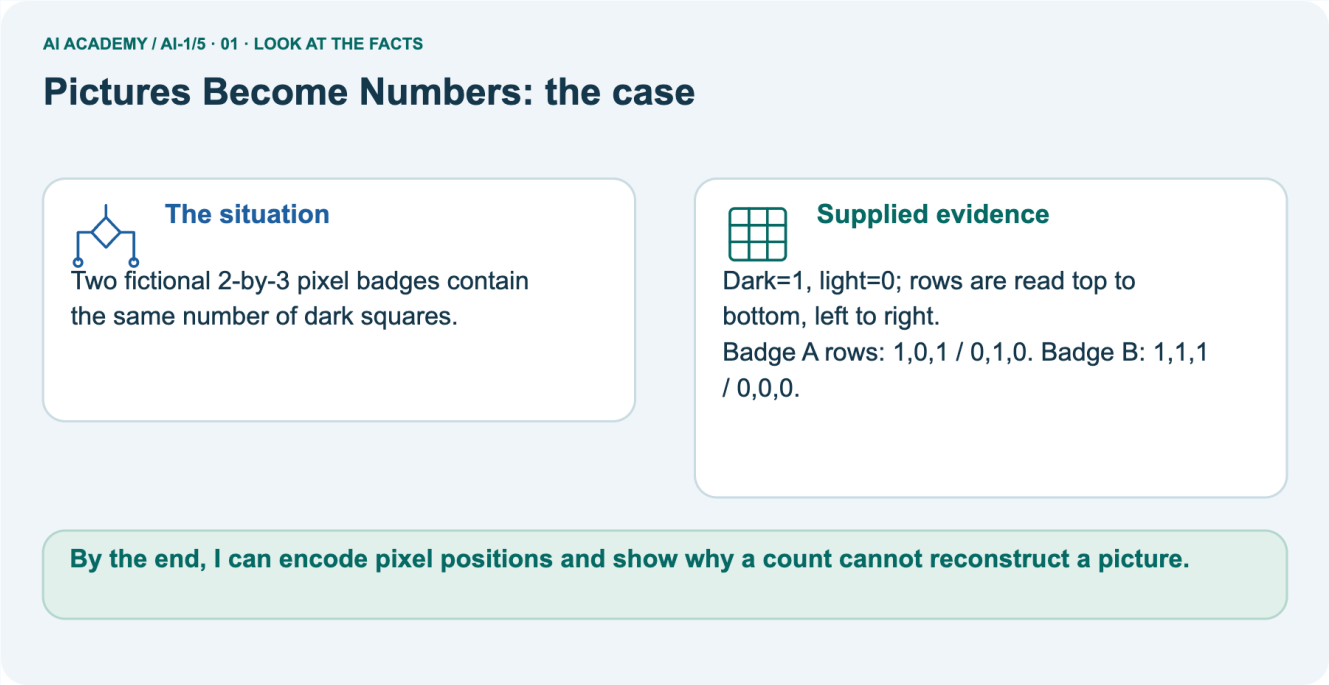
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Pictures Become Numbers graphic using its labels, deciding evidence and relevant limitation.

01. Pictures Become Numbers: the case

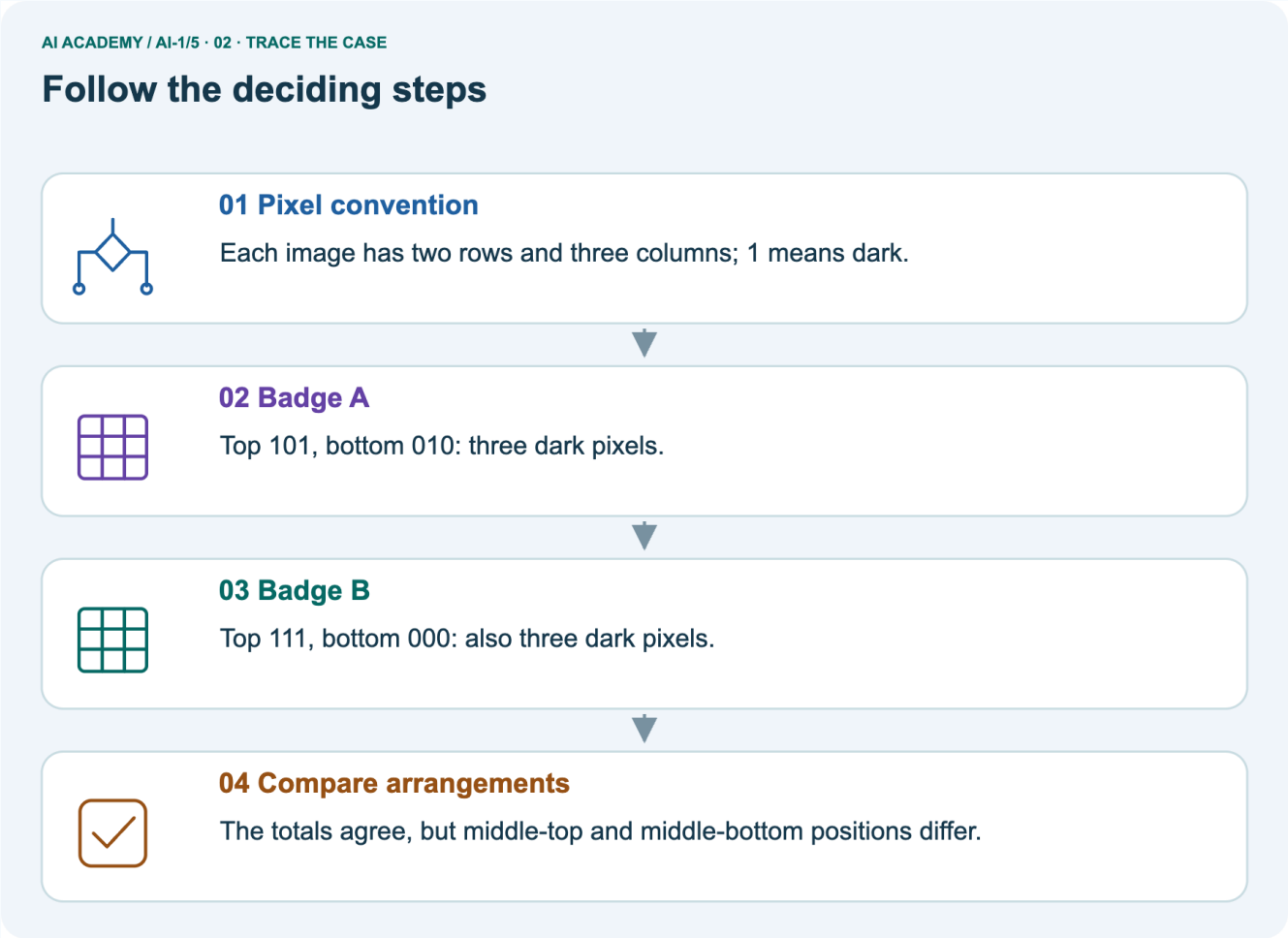


Follow the labelled picture with the matching explanation. Open the vector diagram

Two fictional 2-by-3 **pixel** badges contain the same number of dark squares. Supplied facts: Dark=1, light=0; rows are read top to bottom, left to right. Badge A rows: 1,0,1 / 0,1,0. Badge B: 1,1,1 / 0,0,0.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

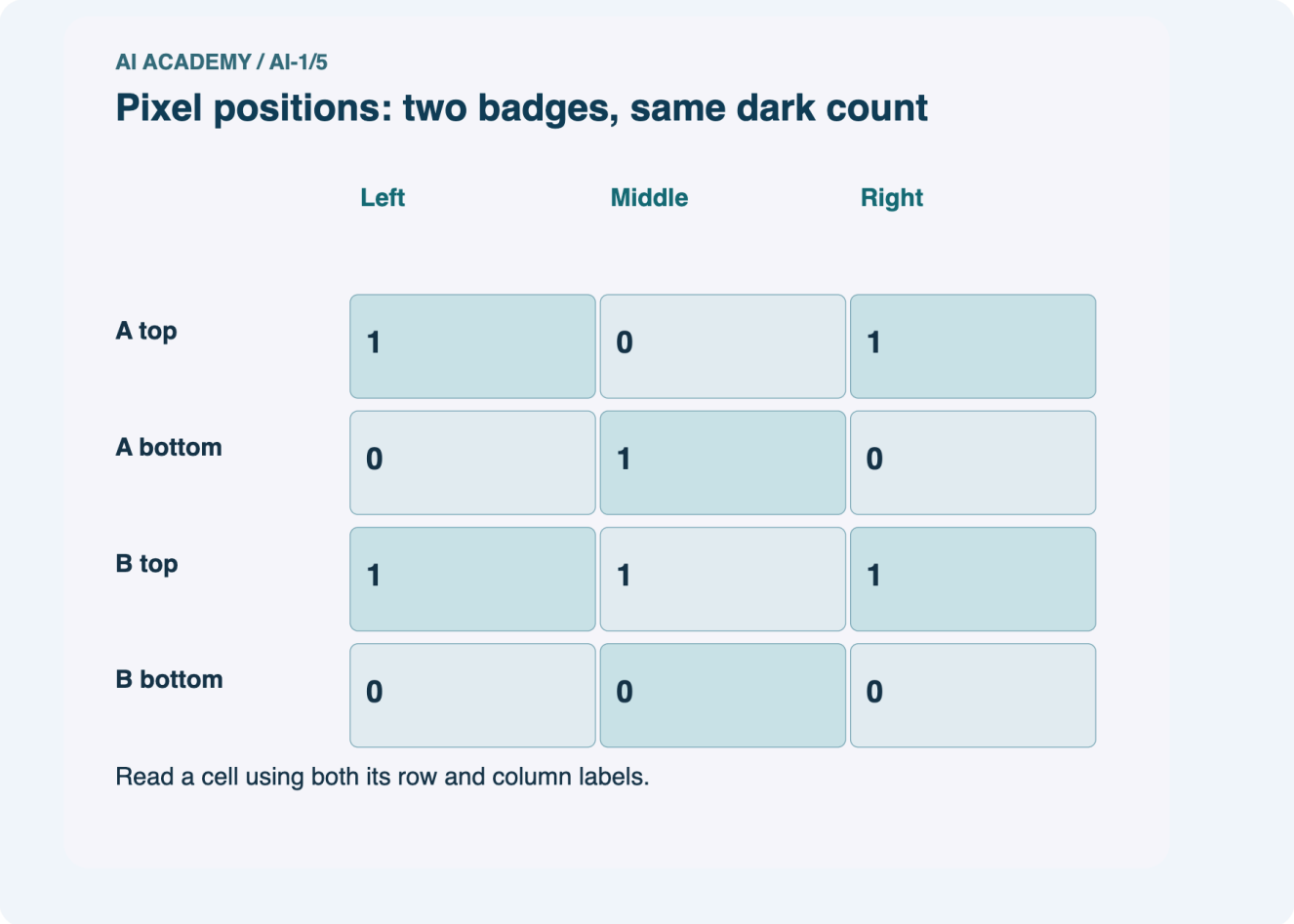


Follow the labelled picture with the matching explanation. Open the vector diagram

Pixel convention: Each image has two rows and three columns; 1 means dark. → Badge A: Top 101, bottom 010: three dark **pixel**s. → Badge B: Top 111, bottom 000: also three dark **pixel**s. → Compare arrangements: The totals agree, but middle-top and middle-bottom positions differ.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Pixel positions: two badges, same dark count

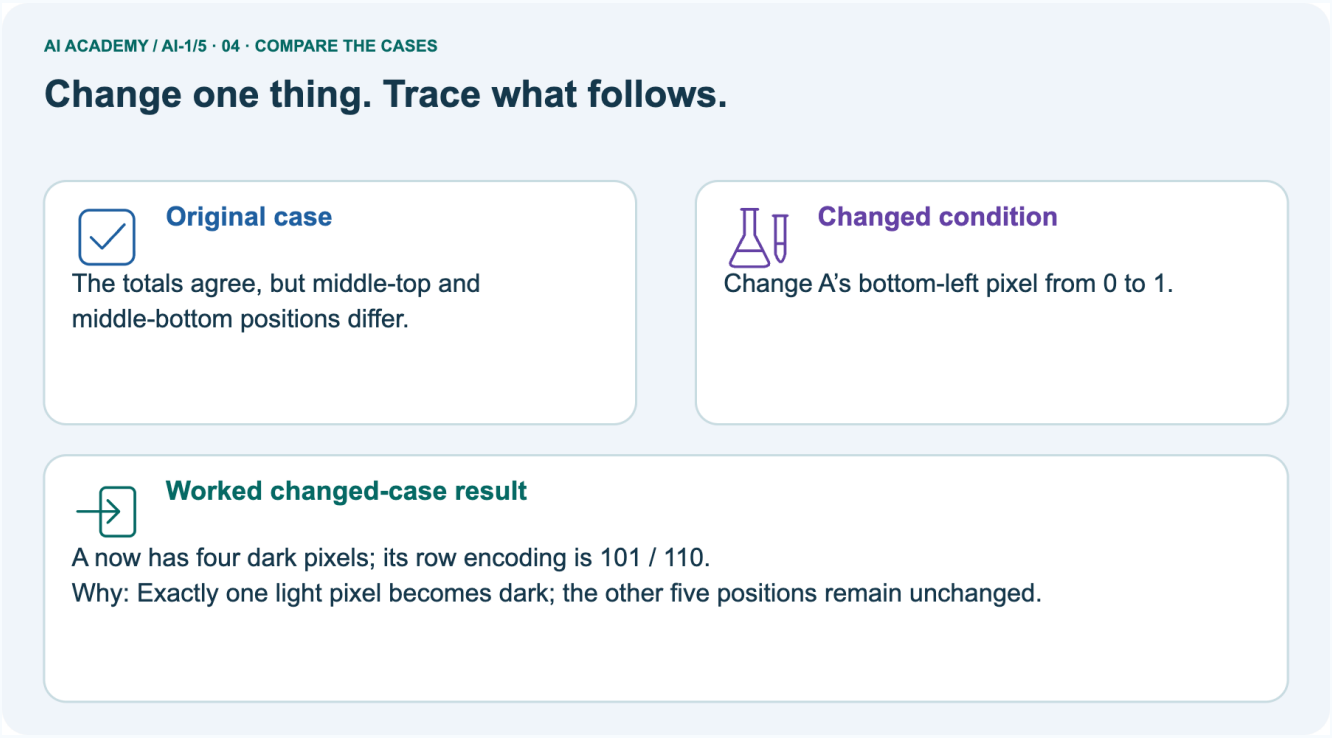


Follow the labelled picture with the matching explanation. Open the vector diagram

Pixel positions: two badges, same dark count. A top: Left = 1, Middle = 0, Right = 1. A bottom: Left = 0, Middle = 1, Right = 0. B top: Left = 1, Middle = 1, Right = 1. B bottom: Left = 0, Middle = 0, Right = 0.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



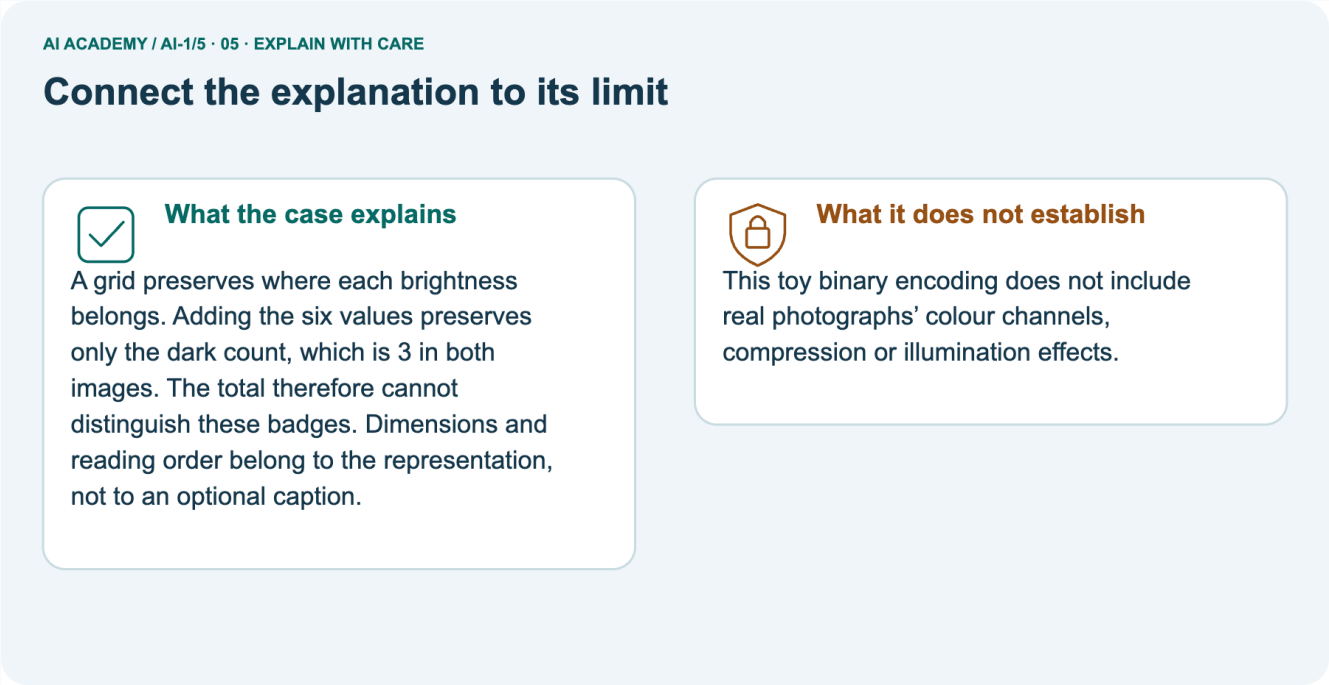
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The totals agree, but middle-top and middle-bottom positions differ. Changed condition: Change A's bottom-left **pixel** from 0 to 1. Expected result: A now has four dark **pixel**s; its row **encoding** is 101 / 110.

Reason: Exactly one light **pixel** becomes dark; the other five positions remain unchanged.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



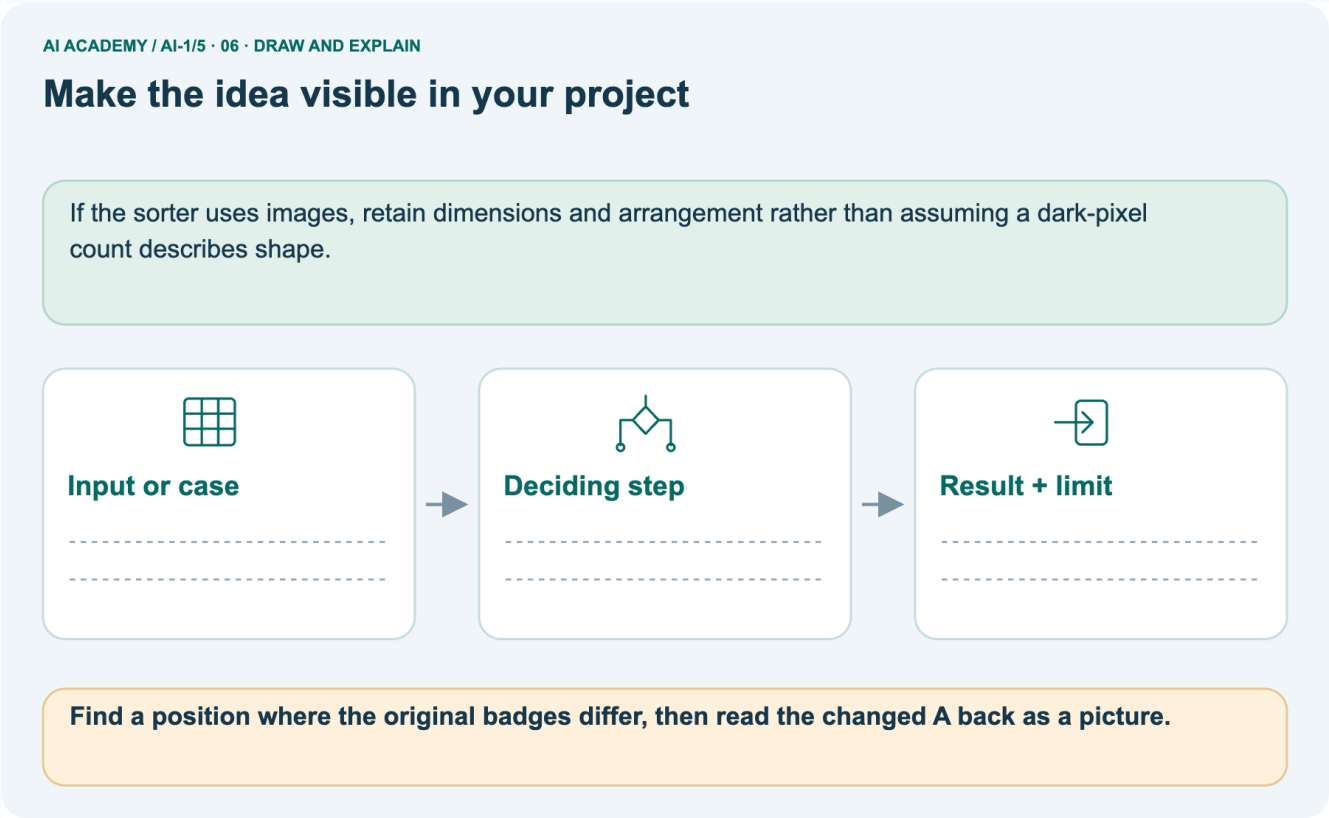
Follow the labelled picture with the matching explanation. Open the vector diagram

A **grid** preserves where each brightness belongs. Adding the six values preserves only the dark count, which is 3 in both images. The total therefore cannot distinguish these badges.

Dimensions and reading order belong to the representation, not to an optional caption. Limit: This toy binary **encoding** does not include real photographs' colour **channel**s, compression or illumination effects.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

If the sorter uses images, retain dimensions and arrangement rather than assuming a dark-**pixel** count describes shape. Draw your **input**, deciding step, result and limit.

Supported question: Find a position where the original badges differ, then read the changed A back as a picture.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.